Meraj Hossain Promit

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Education

University of Dhaka — Dhaka, Bangladesh
B.Sc. in Robotics and Mechatronics Engineering — Aug 2022 – Aug 2026 (Expected)

Sylhet MC College — Sylhet, Bangladesh
Higher Secondary Examination, HSC — July 2019 – December 2021

Publications

Meraj Hossain Promit, Chandak Chakma, Md Jubair Ahmed Sourov, and Sejuti Rahman. "Escape Without a Recipe: Maneuver-Agnostic Mars Rover Recovery from Granular Entrapment." *Space Robotics Workshop, IEEE/RSJ IROS 2026* (poster). Pittsburgh, PA, USA, September 2026.

Experience

Intelis Solution Ltd. — Dhaka, Bangladesh
Embedded Engineer (Part-time) — *July 2026 – Present*

- Building STM32-based autonomous drone platforms for maritime monitoring; developing robust firmware, hardened telemetry, and weather-resistant sensing.

University of Texas Rio Grande Valley — Remote
Research Assistant — *August 2025 – Present*

- Building a digital-twin warehouse benchmark to evaluate Deep RL navigation policies (PPO, SAC, TD3) under realistic perception and localization noise.
- Developing curriculum learning and domain randomization pipelines in NVIDIA Isaac Lab to close the sim-to-real gap on a Clearpath Jackal.

Karthikesh Robotics Private Limited — Chennai, India
ROS Development Intern (Remote) — *June 2025 – Present*

- Developing ROS2-based navigation and localization stack for autonomous mobile robots targeting industrial warehouse automation.
- Implementing sensor fusion algorithms integrating LiDAR and IMU data for improved odometry and SLAM performance.
- Optimizing path planning algorithms using Nav2 stack, reducing computational overhead and improving real-time responsiveness.

Mohakashchari - Dhaka University Rover Team — Dhaka, Bangladesh
Project Lead — *May 2025 – Present*

- Leading cross-functional team of 10+ students to design and build an autonomous Mars rover for international competitions.
- Managing project timeline, budget allocation, and fundraising efforts to secure resources for hardware procurement.
- Overseeing mechanical design and frame integration using CAD tools, ensuring structural integrity for rugged terrain navigation.

MARVEL Lab, University of Dhaka — Dhaka, Bangladesh
Undergraduate Researcher (Supervisor: Dr. Sejuti Rahman) — *April 2025 – Present*

- Lead author on deep RL controllers that recover Mars rover mobility in deformable granular regolith, where classical control fails under slip; paper accepted to the Space Robotics Workshop at IEEE/RSJ IROS 2026 (poster).
- Designed yaw-projected reward shaping with entrapment-gated slip penalties as a safety-style constraint, and built large-scale, multi-GPU training and evaluation pipelines that accelerated policy learning by 3-4x.
- Built training and evaluation pipelines in NVIDIA Isaac Lab with domain randomization and curriculum learning to close the sim-to-real gap.

IEEE Robotics & Automation Society, University of Dhaka Dhaka, Bangladesh
Chair *May 2025 – Present*

- Leading 50+ member chapter, coordinating technical workshops, guest lectures, and robotics competitions.
- Establishing partnerships with industry leaders and research institutions to facilitate knowledge exchange and collaborative projects.

IEEE Robotics & Automation Society, University of Dhaka Dhaka, Bangladesh
Student Activity Secretary *Apr 2023 – May 2024*

- Coordinated 15+ technical workshops and seminars on robotics, automation, and embedded systems for 200+ participants.
- Managed logistics, budgets, and promotional campaigns for chapter events, increasing student engagement by 40%.

North South University Dhaka, Bangladesh
Private Research Student *Jan 2024 – Dec 2024*

- Conducted research on melanoma segmentation using pseudo annotations and deep learning on the ISIC 2017 dataset.
- Implemented preprocessing pipeline including DullRazor hair removal algorithm, improving lesion detection accuracy by 12%.
- Trained U-Net and DeepLabv3+ models for semantic segmentation, achieving Dice coefficient of 0.87+ on validation set.
- Supervised by Dr. Mahdy Rahman Chowdhury, Associate Professor, Department of ECE, NSU.

Skills

Programming Languages: C/C++, Python, CUDA, Bash
Machine Learning & AI: PyTorch, TensorFlow, Deep Reinforcement Learning, OpenCV, Computer Vision
Robotics Technologies and Tools: ROS2, Gazebo, Git, Fusion 360, Raspberry Pi, STM32, ANSYS
Operating Systems & Environments: Linux (Ubuntu, Arch, Kali), Command Line, Docker
Hardware Skills: Soldering, Welding, CNC Milling/Turning, Laser Cutting, Hardware Integration, PCB Design

Awards & Achievements

Competitions and Olympiads:

- 🏆 **8th Place, DU AI Challenge** (August 2025)
  - Computer vision using Deep Learning
- 🏆 **4th Place, Bangladesh Olympiad on Astronomy & Astrophysics** (January 2021)
  - National Winner and Extended Camper

Academic Performance:

- 🎓 **Higher Secondary Examination**, Sylhet Education Board (2021)
  - Ranked 62nd on the Sylhet Education Board

Projects

STM32 Quadcopter Flight Controller | *C/C++, STM32F103, Sensor Fusion, PID*

- Built a bare-metal flight controller on an STM32F103, closing six control loops at 250 Hz with no RTOS and no flight stack; flown and upgraded across undergraduate years.
- Fused gyro, accelerometer, magnetometer, barometer, and GPS via a complementary filter into an attitude and position estimate feeding a rate-PID cascade and motor mixer.

Manipulator Kinematics and Dynamics from Scratch | *ROS2, NumPy, PyBullet, Numerical IK*

- Wrote a full kinematics and dynamics stack from first principles in NumPy, validated against official URDFs for a 6-DOF UR5e and a 7-DOF KUKA iiwa; model and URDF agree to 1.5e-7.
- Caught a DH sign error in the UR5e table that left the model 1.998 m off; implemented numerical IK with damped least squares and Cartesian paths via quintic blending and quaternion slerp.

Melanoma Segmentation Using Pseudo Annotations | *Python, PyTorch, U-Net, OpenCV*

- Developed deep learning pipeline for melanoma segmentation on ISIC 2017 dataset (2,000+ dermoscopic images).
- Implemented preprocessing pipeline with DullRazor hair removal and data augmentation, improving model robustness.
- Trained U-Net architecture with pseudo-label refinement, achieving Dice score of 0.87 and IoU of 0.79 on test set.

WarehouseBenchmark: Deep RL Navigation | *PyTorch, Isaac Lab, ROS2, PPO/SAC/TD3*

- Built a 31.8×54 m digital-twin warehouse benchmarking PPO, SAC, and TD3 for autonomous navigation on a Clearpath Jackal under realistic LiDAR perception and localization noise.
- Applied curriculum learning and domain randomization to close the sim-to-real gap; integrated with a ROS2 navigation stack.

Relevant Coursework

Core Robotics: Introduction to Robotics, Artificial Intelligence, Intelligent Systems & Robotics, Control Systems Design, Intro to Mechatronics Engineering, Advanced Mechatronics Engineering, Digital Logic Circuits and Microprocessors, Micro-controller and Programmable Logic Controller, Manufacturing Process with CNC Programming
Software: Data Structures and Algorithms, Object-Oriented Programming

Math and AI: Linear Algebra, Calculus, Mathematical Analysis for Engineering (Numerical Analysis, Laplace & Fourier Transformation), Statistics for Engineers, Multivariable Vector Calculus, Differential Equations and Coordinate Geometry

Hobbies & Interests

Stargazing, Cycling, Traveling, Singing, Creative Writing, Networking, Ethical Hacking, Chess